P.6 Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus P.6 (30-minute plan) · Student sheet

I can…

- find exact values of the trigonometric functions in radians.
- use the period and the even-odd properties to evaluate trigonometric functions.
- find the amplitude and period of a sinusoidal function and write its equation.

Vocabulary · fill in the missing word

Periodic function — a function f with a _____ number p so that f(x + p) = f(x) for every x in its domain

Period — the _____ such number p

Amplitude — |A| for y = _____ = A cos(ωx)

Part 1 · Notes and examples (fill in each note, then work the example)

Domain and range: sine and cosine are defined for all real numbers and have range [−1, 1]; tan x is undefined at odd integer multiples of _____.

Example 1

Find the exact value.

a. $\sin(\frac{5\pi}{6})$

b. $\cos(\frac{7\pi}{6})$

c. $\tan(\frac{3\pi}{4})$

Periods: sine, cosine, secant, and cosecant have period 2π; tangent and cotangent have period _____.

Even-odd: cosine and secant are even; sine, tangent, cosecant, and cotangent are _____.

Sinusoids: y = A sin(ωx) and y = A cos(ωx), ω > 0, have amplitude |A| and period _____.

Example 2

Find the amplitude and period.

a. $g(x) = (\frac{1}{2})\cos(\pi x)$

b. $f(x) = -2\cos((\frac{3}{2})x)$

Example 3

a. Write a sine function with amplitude 2 and period π.

b. Write a cosine function with amplitude 3 and period 4π.

Part 2 · Partner practice

Partner practice

Find $\sin(-\frac{\pi}{2})$ and $\cos(\frac{13\pi}{6})$.